

Department of Mathematics "F. Casorati",
University of Pavia, Via Ferrata 5,
Pavia, 27100, Italy

in [Sara Cambiaghi](#)
✉ sara.cambiaghi01@universitadipavia.it
🐙 [github/SaraCambiaghi](https://github.com/SaraCambiaghi)

SARA CAMBIAGHI

CURRENT POSITION

PhD student in Computational Mathematics, Learning and data science at University of Pavia, 2023–Present.

My research project is *Optimization and machine learning for the efficiency of public hospital services*. The project focuses on the study of combinatorial optimization methods and data science models to support decision-making in the management of healthcare services, with the goal of improving the efficiency of resource utilization. To achieve this objective, the impact of alternative strategies will be evaluated from the perspective of multiple stakeholders: patients (service quality and fairness), healthcare professionals (workload distribution), and hospital management (resource utilization and throughput).

EDUCATION

M.Sc. in Mathematics, University of Milano-Bicocca, 2023.

Dissertation: *Selecting Subgame Perfect Nash Equilibria in Stackelberg Games*.

Final grade: 110L/110.

B.Sc. in Mathematics, University of Milano-Bicocca, 2021.

Dissertation: *Benford's Law*.

Final grade: 109/110.

PUBLICATIONS

Sara Cambiaghi, Davide Duma. (2026). *Multi-objective outpatient scheduling via learning-based adaptive evolutionary search*. Computers & Operations Research.

Accepted papers

Alice Daldossi, Sara Cambiaghi, Davide Duma, Roberto Aringhieri. (2026). *Prediction accuracy or decision quality? A predict-then-optimize perspective on surgical case assignment and sequencing*. Shaping a Sustainable Future in the Era of Big Data, Proceedings of the 9th AIRO Young Workshop, AIRO Springer Series.

Submitted papers

Ludovico Ambrosi, Chandra Bortolotto, Sara Cambiaghi, Luisa Carone, Davide Duma, Lorenzo Preda. *Optimizing Multi-Stakeholder Fairness in Computed Tomography Appointment Scheduling with Predicted Reporting Duration*.

Sara Cambiaghi, Saverio Basso, Davide Duma, Matteo Salani *A column generation approach to joint master surgical scheduling and case assignment under bed capacity constraints*.

Working papers	Sara Cambiaghi, Davide Duma <i>From Prediction to Optimization: A Literature Review and Comparative Benchmark for Patient Scheduling Under Uncertainty</i> .
TEACHING	Instructional seminars within the teaching of Mathematical Methods, supervised by Professor Davide Duma, for the bachelor's degree in Bioengineering, 2025. Tutorial session on <i>Introduction to MATLAB</i>, supervised by Professor Benedetta Ferrario, for the master's degree in Finance, 2024. Instructional seminars within the teaching of Mathematical Methods, supervised by Professor Davide Duma, for the bachelor's degree in Bioengineering, 2024.
SCIENTIFIC MEETINGS	
Regular talk	A column-generation approach to joint master surgical scheduling and case assignment under bed capacity constraints. 10th AIROYoung Workshop, 2026. Warm-Starting Outpatient Appointment Scheduling Using Historical Solutions. ORAHS, 2025. Operating Room Scheduling under Uncertainty: Point Predictions vs. Conditional Density. ORAHS Doctoral Colloquium, 2025. Appointment scheduling of CT-scans in the presence of emergency arrivals. 8th AIROYoung Workshop, 2024. Predictive Analytics for Exam Duration and Reporting Times to Enhance CT Scan Scheduling. ORAHS Doctoral Colloquium, 2024. CT-Scan Outpatient Scheduling under Uncertain Durations and Emergency Arrivals. Mathematics for Artificial Intelligence and Machine Learning, 2023.
Poster presentation	Distributional Forecast Approaches to Stochastic Optimization in Healthcare Appointment Scheduling. 3rd Workshop of the UMI Group on Mathematics for Artificial Intelligence and Machine Learning, 2025. Scheduling CT Scans Under Uncertainty via Memory Augmented NSGA II. Spring Workshop COMPMAT, 2025. Comparing machine learning models in a predict-then-optimize framework for outpatient CT-scan scheduling. CompMat 2024 Spring Workshop, 2024.

Stochastic optimization approach for scheduling CT scans and reports. ORAHS, 2024.

AWARDS Second place at the Steve Gallivan Prize Award. ORAHS, 2024.

EXTRA Organization of the 9th AIROYoung Workshop, 2025.
ACTIVITIES Visiting period at USI, Lugano, Switzerland. *September 2025 – March 2026.*

Pavia (PV), Italy
July 31, 2026

Sara Cambiaghi

A handwritten signature in black ink, reading "Sara Cambiaghi", written over a horizontal line.

Signature